



MASTER TUTOR SCRIPT: MHF4U UNIT 5 - TRIG IDENTITIES & EQUATIONS Student: Rigden Total Slides: 24

HOW TO USE THIS SCRIPT: Keep this text file open on your second monitor. The HTML presentation file is completely clean and student-safe. Guide Rigden through the interactive slides using the prompts, questions, and expected answers below.

SLIDE 1: Title Slide

[SAY] Welcome back, Rigden! We've mastered graphing trigonometric waves and using radians. Today, we enter Unit 5. [SAY] This unit is heavy on algebra. We are going to combine the factoring skills you learned in Unit 2 with the trigonometry you learned in Unit 4. Welcome to Trigonometric Identities and Equations.

SLIDE 2: Today's Roadmap

[SAY] Here is our game plan for today's session. [SAY] First, we'll establish what an identity actually is. Second, we'll learn the Compound and Double Angle formulas. Third, we will prove complex identities. And finally, we will solve high-level trigonometric equations.

SLIDE 3: Equation vs. Identity

[SAY] Before we do any math, we need to clear up a massive misconception. What is the difference between an Equation and an Identity? [SAY] Look at the left: $\sin(x) = 1/2$. This is an equation. It is only true for specific values of x , like $\pi/6$ or $5\pi/6$. Our goal is to SOLVE for x . [SAY] Look at the right: $\sin^2(x) + \cos^2(x) = 1$. This is an identity. It is true for literally ANY number you plug in. 5, 20, 1 million—it will always equal 1. Our goal here isn't to solve for x , but to PROVE that the left side equals the right side.

SLIDE 4: Warm-up: The Grade 11 Foundations

[SAY] To do Grade 12 proofs, you need to remember your Grade 11 toolbox. [ASK] What is the Quotient Identity? How can we rewrite $\tan(x)$? [EXPECT] $\sin(x) / \cos(x)$. [ASK] Perfect. What about the Reciprocal Identities? What is $\csc(x)$, $\sec(x)$, and $\cot(x)$? [EXPECT] $\csc$ is $1/\sin$. $\sec$ is $1/\cos$. $\cot$ is $1/\tan$ or $\cos/\sin$. [SAY] Excellent. And finally, the Pythagorean Identity: $\sin^2(x) + \cos^2(x) = 1$. You will use this in almost every proof.

SLIDE 5: Equivalent Trig Expressions

[SAY] Let's look at the periodic nature of waves. Because sine and cosine repeat forever, we can shift them and get the exact same value. [SAY] If you have an angle ' x ', and you add 2π (a full circle), you end up at the exact same spot! [ASK] Look at the equations. If $\sin(x) = 0.5$, what is $\sin(x + 2\pi)$? [EXPECT] It's also 0.5. [SAY] Exactly. You can always add or subtract 2π from an angle inside a sine or cosine function without changing the answer.

SLIDE 6: Co-function Identities

[SAY] Now, what is the relationship between the Sine wave and the Cosine wave? [ASK] If you look at their graphs, they are the exact same shape. How far shifted is the Cosine wave from the Sine

wave? [EXPECT] 90 degrees, or $\pi/2$. [SAY] Spot on. This gives us the Co-function identities. Cosine is just Sine shifted to the left by $\pi/2$. And Sine is Cosine shifted to the right by $\pi/2$.

SLIDE 7: The Compound Angle Formulas

[SAY] Here is where Grade 12 really begins. What if you want to find the exact value of $\cos(7\pi/12)$? [SAY] $7\pi/12$ is not on our special triangles. We only know $\pi/3$, $\pi/4$, and $\pi/6$. [SAY] But, what if we could break $7\pi/12$ into two angles we DO know? Like $3\pi/12 + 4\pi/12$ (which simplifies to $\pi/4 + \pi/3$). [SAY] To do this, we need the Compound Angle Formulas.

SLIDE 8: The Cosine Compound Formulas

[SAY] First up, Cosine. The formula for $\cos(a + b)$ is $\cos(a)\cos(b) - \sin(a)\sin(b)$. [TIP] Point out the memory trick. "Cosine is cliquey and contrary. The cosines group together, the sines group together, and the sign in the middle is the OPPOSITE of what you started with." [ASK] If we have $\cos(a - b)$, what will the sign in the middle be? [EXPECT] A plus sign (+).

SLIDE 9: The Sine Compound Formulas

[SAY] Next is Sine. The formula for $\sin(a + b)$ is $\sin(a)\cos(b) + \cos(a)\sin(b)$. [TIP] Point out the memory trick. "Sine is social and same. Sine mixes with cosine, and the sign in the middle stays EXACTLY the SAME." [ASK] So for $\sin(a - b)$, the terms are mixed, and the sign in the middle is...? [EXPECT] A minus sign (-).

SLIDE 10: The Tangent Compound Formulas

[SAY] Tangent is a bit messier because it's a fraction. [SAY] $\tan(a + b) = (\tan a + \tan b) / (1 - \tan a \tan b)$. [SAY] The sign on the TOP matches the original operation. The sign on the BOTTOM is the opposite.

SLIDE 11: Practice: Evaluating Exact Values

[SAY] Let's test this out. Find the exact value of $\sin(5\pi/12)$. [SAY] First, break $5\pi/12$ into two special angles. $2\pi/12 + 3\pi/12$ works perfectly, which simplifies to $\pi/6 + \pi/4$. [ASK] We are doing $\sin(a + b)$. Sine is "social and same". Set up the formula for me. [EXPECT] $\sin(\pi/6)\cos(\pi/4) + \cos(\pi/6)\sin(\pi/4)$. [SAY] Exactly! Now you just plug in the exact values from your special triangles and simplify the fractions.

SLIDE 12: Double Angle Formulas (Sine & Tangent)

[SAY] What if 'a' and 'b' are the exact same angle? What is $\sin(x + x)$? That's $\sin(2x)$. [SAY] If you plug 'x' and 'x' into the compound formula, it simplifies beautifully. [SAY] $\sin(2x)$ always equals $2 \sin(x) \cos(x)$. [SAY] $\tan(2x)$ equals $(2 \tan x) / (1 - \tan^2 x)$.

SLIDE 13: The 3 Faces of Cosine Double Angle

[SAY] Cosine is special. $\cos(2x)$ has THREE different versions. [SAY] The main one is $\cos^2(x) - \sin^2(x)$. [SAY] But if we use the Pythagorean identity ($\sin^2 + \cos^2 = 1$) to swap out terms, we get two alternates. [ASK] Look at the three versions. When proving an identity, why might it be super useful to have a version that ONLY uses sine, or ONLY uses cosine? [EXPECT] It helps you match the other side of the equation if the other side only has sine or cosine!

SLIDE 14: Practice: Double Angle Expansion

[SAY] Let's see if you can spot a double angle in reverse. [SAY] Simplify: $2 \sin(\pi/8) \cos(\pi/8)$. [ASK] This matches the pattern $2 \sin(x) \cos(x)$. What is 'x' in this case? [EXPECT] $\pi/8$. [SAY] Right. So we

can condense this whole thing back into $\sin(2x)$. What is $2 * (\pi/8)$? [EXPECT] $2\pi/8$, which is $\pi/4$. [SAY] Spot on. And we know $\sin(\pi/4)$ is $1/\sqrt{2}$. You just evaluated a complex expression in seconds!

SLIDE 15: Proving Identities: The Golden Rules

[SAY] Now we enter the proof section. The goal is to prove LHS = RHS. [SAY] Here are the Golden Rules of proofs. [SAY] 1. NEVER cross the line. You cannot move terms from left to right. You work on one side independently. [SAY] 2. Start with the messiest, ugliest side. It's easier to simplify a mess than to complicate something simple. [SAY] 3. When in doubt, convert everything to sine and cosine.

SLIDE 16: Identity Proof: Example 1

[SAY] Let's prove: $\tan(x) + \cot(x) = \sec(x)\csc(x)$. [SAY] Step 1: The left side is messier. Let's convert tan and cot into sines and cosines. [ASK] What does tan + cot become? [EXPECT] $(\sin/\cos) + (\cos/\sin)$. [SAY] Great. Step 2: Get a common denominator to add the fractions. It becomes $(\sin^2 + \cos^2) / (\sin \cos)$. [ASK] Step 3: Look at the numerator. $\sin^2 + \cos^2$. What identity is that? [EXPECT] Pythagorean! It equals 1. [SAY] Exactly! So we have $1 / (\sin \cos)$. Which splits into $(1/\cos) * (1/\sin)$. Which is $\sec(x)\csc(x)$. LHS = RHS!

SLIDE 17: Identity Proof: Example 2 (Double Angles)

[SAY] Let's look at one with double angles. Prove: $(\sin 2x) / (1 + \cos 2x) = \tan x$. [SAY] The left side is messier. We need to get rid of the $2x$. [ASK] What is the only formula for $\sin(2x)$? [EXPECT] $2 \sin(x) \cos(x)$. [SAY] Good. Now look at the denominator: $1 + \cos(2x)$. We have 3 choices for $\cos(2x)$. We want to cancel out that '1'. Which of the 3 formulas has a '-1' in it? [EXPECT] $2 \cos^2(x) - 1$. [SAY] Brilliant! Plug that in. The 1 and -1 cancel. You are left with $(2 \sin x \cos x) / (2 \cos^2 x)$. The 2s cancel, one cos cancels, and you are left with $\sin/\cos$, which is $\tan(x)$.

SLIDE 18: Your Turn! Prove the Identity

[SAY] Grab your paper. I want you to prove: $\cos(x) / (1 - \sin x) = (1 + \sin x) / \cos(x)$. [TIP] If Rigden gets stuck: Tell him to multiply the Left Side by the conjugate $(1 + \sin x)$ on both the top and the bottom! This will create a difference of squares on the bottom. [SAY] Once you get $1 - \sin^2(x)$ on the bottom, use the Pythagorean identity to turn it into $\cos^2(x)$. Then cancel!

SLIDE 19: Solving Trig Equations: Linear Review

[SAY] Let's shift gears from identities back to solving equations. We want to find 'x'. [SAY] Solve: $2 \sin(x) - 1 = 0$ between 0 and 2π . [ASK] How do we isolate $\sin(x)$? [EXPECT] Move the 1 over, divide by 2. $\sin(x) = 1/2$. [SAY] Exactly. Now we use Unit 4 skills. Sine is positive in Q1 and Q2. The reference angle for $1/2$ is $\pi/6$. So $x = \pi/6$ and $5\pi/6$.

SLIDE 20: Solving Trig Equations: Quadratic

[SAY] What if it's a quadratic? $2 \sin^2(x) - \sin(x) - 1 = 0$. [SAY] Treat $\sin(x)$ like a giant 'A'. This is just $2A^2 - A - 1 = 0$. [ASK] We factor this just like Grade 10 math. What does it factor into? [EXPECT] $(2A + 1)(A - 1) = 0$. [SAY] Perfect. So $(2 \sin x + 1)(\sin x - 1) = 0$. This splits into TWO separate linear equations. $\sin(x) = -1/2$, and $\sin(x) = 1$. You solve both to get all your answers!

SLIDE 21: Worked Example: Quadratic Trig Eq

[SAY] Let's look at the board to see the full solution to the quadratic we just factored. [SAY] For $\sin(x) = -1/2$, sine is negative in Q3 and Q4. Ref angle is $\pi/6$. Answers: $7\pi/6$ and $11\pi/6$. [SAY] For

$\sin(x) = 1$, where on the unit circle is the y -value exactly 1? Right at the top! So $x = \pi/2$. [SAY] So this equation has 3 valid answers in the domain.

SLIDE 22: Solving Trig Equations with Double Angles

[SAY] Here is the ultimate trap. Solve: $\cos(2x) + \cos(x) = 0$. [SAY] You CANNOT factor this right now because the angles don't match. One is ' $2x$ ', the other is ' x '. [ASK] We need to swap out $\cos(2x)$. Which of the 3 formulas should we use so the whole equation is in terms of cosine? [EXPECT] $2\cos^2(x) - 1$. [SAY] Yes! Swap it in. Now you have $2\cos^2(x) + \cos(x) - 1 = 0$. It's a regular quadratic! Factor it and solve.

SLIDE 23: The "Extraneous Root" Trap

[SAY] One last warning for equations. [SAY] Look at $\sin(x)\cos(x) = \sin(x)$. Your brain probably wants to divide both sides by $\sin(x)$ to get $\cos(x) = 1$. [SAY] NEVER divide an equation by a variable expression! If you do, you just deleted half your answers. [SAY] Instead, move everything to one side: $\sin(x)\cos(x) - \sin(x) = 0$. Then common factor out the $\sin(x)$. [SAY] $\sin(x) [\cos(x) - 1] = 0$. This gives you way more answers!

SLIDE 24: Unit 5 Consolidation

[SAY] You did an incredible job today, Rigden. We covered compound angles, double angles, proofs, and quadratics. Let's do a rapid-fire exit ticket to wrap up. [ASK] 1. What is the difference between an identity and an equation? (Expected: Identity is true for all x , equation is solved for specific x). [ASK] 2. Expand $\sin(a - b)$. (Expected: $\sin a \cos b - \cos a \sin b$). [ASK] 3. What is the golden rule of proving identities? (Expected: Work on one side only, don't cross the equals sign). [SAY] Perfect. You are ready to tackle the Unit 5 practice problems next!